



Impact of Moringa oleifera extracted and amino acid on growth and quality of olive tree (Olea europaea)

***Al Ziyituni, O. H.**

***Faculty of Veterinary Medicine and Agricultural Sciences -AlAejaylat, University of Zawia**

h.alzyituni@zu.edu.ly

<https://orcid.org/0009-0008-2016-2898>

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ABSTRACT

During the 2025 season, an experiment was conducted in the nursery of the private farm Alaejaylat in Libya to assess the impact of extracted *Moringa oleifera* and amino acids on the development and quality of olive trees of the "Koroneiki" variety. Ten treatments (control, Moringa extracts at 4, 8, and 12 g/l), amino acids at 50, 75, and 100 mg/l, and mixtures MLAE 4 g/l + Amino acid 50 mg/l, MLAE 8 g/l + Amino acid 75 mg/l, and MLAE 12 g/l + Amino acid 100 mg/l) were used in three replicates. Plant height, leaf count, leaf area index, and total chlorophyll are examples of vegetative growth; fruit length, fruit diameter, fruit weight, flesh weight, and yield yield/tree are examples of yield and yield components; and nitrogen, phosphate, potassium, protein, total carbohydrates, and oil content are examples of chemical composition. The plant height, number of leaves, leaf area index, total chlorophyll, fruit length, fruit diameter, fruit weight, flesh weight, yield yield/tree, nitrogen, phosphate, potassium, protein, total carbohydrates, and oil content were all higher in the mixture MLAE 12 g/l + Amino acid 100 mg/l. In comparison to the control treatment, which showed the lowest mean values of plant height, number of leaves, leaf area index, and total chlorophyll, fruit length, fruit diameter, fruit weight, flesh weight, yield yield/tree, nitrogen, phosphate, potassium, protein, total carbohydrates, and oil content, the treatment of MLAE 12 g/l and amino acid 100 mg/l came next. In conclusion, spraying olive trees with moringa leaf extract increased the amount of certain chemicals and minerals in their leaves. Moringa leaf extract is thought to be an environmentally friendly method of improving plant growth. Ultimately, more thorough and comparative research is required to fully elucidate the specific roles and optimal applications of individual amino acids in plant growth, which will be crucial for developing effective biostimulants and advancing agricultural practices. In conclusion, spraying olive trees with moringa leaf extract had a significant role in increasing the content of the leaves of some compounds and nutrients. Moringa extract on plants is considered an environmentally friendly approach to improve plant growth.

Keywords: Olive tree (*Olea europaea*), moringa leaf extract, amino acids, vegetative growth, yield parameters, chemical composition.

INTRODUCTION

Olea europaea L., the olive tree, is one of the most widely distributed fruit plants in the world and one of the oldest agricultural products with substantial cultural and commercial value in the Mediterranean Basin. Olive trees, which belong to the Oleaceae family, are evergreen and come in a number of kinds that are used for pickling and oil extraction. Olives also thrive in a range of soil types and growing conditions. Based on data from the Food and Agriculture Organization (**FAOSTAT, 2021**). Olive has a reputation as "the scent of the soft gold" because of its great cultural, social, and economic values. With 10 million hectares under cultivation, olives are currently the most widely grown oleiferous tree species worldwide (**FAO, 2019**). Olive trees are known for their remarkable capacity for morphological and physiological adaptability, despite the fact that they frequently experience drought throughout the warm season in their climate (**Mechri et al., 2020**). Over 90% of the world's olive output comes from Mediterranean countries (**Manetsberger et al., 2023**), whereas 179 countries consume table olives and olive oil (**Cardoni and Mercado-Blanco, 2023**). The significant demand for olive products worldwide and localized production highlight the crop's economic, social, and environmental significance in producing nations. Olive performance is negatively impacted by climate change (**Dias et al., 2021**), despite being one of the crops most suited to the Mediterranean region's climate (**Kaniewski et al., 2023**), which is marked by warm, dry summers with high levels of solar radiation (**Arenas-Castro et al., 2020**).

Amino acids and other biostimulants provide steady, natural ways to increase growth, yield, and tolerance to outside influences. They accomplish this by boosting plant defense systems and lessening the effects of abiotic stress. Amino acids have gained attention as biostimulants due to their affordability, ease of use, and environmental friendliness. They also provide steady, natural solutions to increase crop productivity and tolerance to external stressors (**Henderson et al., 2025**). **Zrig et al. (2023)** have shed important light on how amino acids affect hormone signaling and metabolic adaptation in olive trees exposed to salinity. In olive trees, amino acids act as building blocks for the production of osmoprotectants and stress-responsive metabolites. They take part in the creation of secondary metabolites, polyamines, and compatible solutes that are involved in stress tolerance processes (**Liu et al., 2022** and **El-Esawi and Ali, 2022**). It has been demonstrated that plants under salinity stress accumulate amino acids such proline and glycine betaine. By stabilizing cellular structures, these substances function as osmoprotectants, assisting cells in maintaining turgor pressure and avoiding ion imbalance. According to recent research, proline administration significantly reduces oxidative damage via regulating ion transport and osmotic adjustment (**Ghars et al., 2022**). By building up in plant cells, amino acids aid in osmotic adjustment, preserving turgor pressure and water absorption in saline environments. Among the most important amino acids linked to osmotic regulation in olive trees under salt stress are proline, glycine betaine, and arginine (**Orlandi et al., 2021; Hassan and Ali 2022**). Plant hormones like auxins are derived from amino acids like tryptophan. According to studies, applying tryptophan topically raises auxin levels, which enhances root growth and improves the absorption of nutrients and water. Even in saline conditions, this hormonal regulation supports vegetative development and yield (**Mosleh and Soltani, 2023**). Hormonal signaling mechanisms involved in plant stress responses are influenced by

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amino acids. In olive trees, amino acid mixes control the production, perception, and signaling of stress-related hormones such as jasmonates, cytokinins, and abscisic acid (ABA). According to **Fariduddin *et al.* (2021); Khan and Müller, (2022)**, this modulation aids in coordinating adaptive responses to salt stress. Even in stressful situations, amino acid treatments boost fruit size and weight by improving cell expansion and nutrient transfer (**Di-Vaio *et al.*, 2024**).

Plant extracts have been widely used in the modern era as a supplement or substitute for artificial growth regulators and fertilizers. The development and productivity of many plant species can be improved and enhanced by these plant extracts. The moringa (*Moringa oleifera*) leaf extract in these extracts is a rich source of vitamins and contains significant environmentally friendly biostimulants that significantly contribute to many aspects of growth and development. Along with amino acids, organic acids, steroids, natural antioxidants including flavonoids, ascorbates, and phenols, and some plant hormones like zeatin, moringa leaf extracts also contain significant concentrations of carbohydrates, proteins, and minerals (**Udofia *et al.*, 2020; Zulfiqar *et al.*, 2020**). Antioxidants and antimicrobial properties are abundant in moringa leaf extract (**Kumar *et al.*, 2012; Vongsak *et al.*, 2012**). On the other hand, foliar spraying moringa leaf extract reduced the quantity of unmarketable fruits while increasing the yield and marketable fruit (**Nasira *et al.*, 2016**). According to **Abbassy *et al.* (2020)**, moringa extract enhances growth and yield and is high in proteins, lipids, carbs, minerals, vitamins, and amino acids. According to **Hassan *et al.* (2019)**, applying moringa extract at 0, 2, and 4% to olive trees cv. "Picual" enhanced the total chlorophyll content of the leaves.

This study's primary goal is to find out how *Moringa oleifera* and amino acids impact the vegetative growth, yield, yield components, and chemical composition of "Koroneiki" variety olive trees in the 2025 growing season.

MATERIALS AND METHODS

Experiment was conducted in the nursery of the private farm Alaejaylat in Libya to assess the impact of extracted *Moringa oleifera* and amino acids on the development and quality of olive trees of the "Koroneiki" variety. Ten treatments and three duplicates were used in the Randomized Complete Block Design (RCBD) experiment. The arrangement of the treatments is as follows:

- Control
- MLAE 4 g/l
- MLAE 8 g/l
- MLAE 12 g/l
- Amino acid 50 mg/l
- Amino acid 75 mg/l
- Amino acid 100 mg/l
- MLAE 4 g/l + Amino acid 50 mg/l
- MLAE 8 g/l + Amino acid 75 mg/l
- MLAE 12 g/l + Amino acid 100 mg/l

Preparation moringa leaf extracts:

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Before the investigation, the nutritional makeup of the Moringa leaf powder (MLP) was examined (**Table 1**). Several extractions were carried out with various solvents. Until the extracts were colorless, moringa powder was extracted several times.

Table (1): Chemical composition of dry *Moringa oleifera* leaf powder

Component	Values	Unit
N	4.49	%
P	0.36	%
K	1.62	%
C	44.28	%
Ca	2.71	%
Mg	0.51	%
Na	993.7	mg/kg
Zn	24	mg/kg
Cu	8.5	mg/kg
Mn	57	mg/kg
Fe	842	mg/kg
Al	795	mg/kg
B	18	mg/kg

Data recorded

A) Vegetative growth

- **Shoot length(cm):** the average shoot length was measured in cm
- **Number of leaves/ tree**
- **LA (cm²):** To calculate the average leaf area (cm²), thirty leaves from each tree under treatment were gathered. According to **Yang *et al.* (2019)**, the Digital Leaf Area Metre (LiDE 110, YMJ-C, Top Instrument) was used to scan each sampled leaf.
- **Total chlorophyll (SPAD):** Each plot's total chlorophyll content was determined using a Minolta chlorophyll meter (SPAD-501) at SPA Dunits (**Yadava, 1986**).

B) Yield and yield components (kg/ tree)

- **Fruit length (cm):** A digital caliper was used to measure the sample's length (30 fruits), and the average pulp weight was computed.
- **Fruit diameter (cm):** The sample's diameter (30 fruits) was measured using a digital caliper, and the average pulp weight was computed.
- **Fruit weight (g):** The average fruit weight (g) was determined by weighing the sample of each studied tree.
- **Flesh weight (g):** For every fruit sample, the average weight of flesh per olive fruit (g) was calculated.
- **Fruit Yield (kg/tree):** was measured in kilograms for every tree or duplicate. For every treatment, the average yield (kg)/tree was calculated. The following formula was used to determine fruit yield:

Yield (kg/tree) is calculated by multiplying the average fruit weight by the number of trees.

C) Chemical compositions

- Leaf elements content: Samples of leaves were removed from the treated transplant, cleaned with cotton cloths, weighed fresh, cleansed with tap water and then distilled water, dried, and ground. Following **Cottenie (1980)** instructions, 0.5g of dried materials were digested using H₂SO₄ and H₂O₂. The following mineral content of leaves was ascertained using the extracted:
- According to **(Regni and Proietti, 2019)**, the Microkjeldhal device was used to estimate the nitrogen proportion.
- The flame photometer device was used to measure potassium (**Horneck and Hanson, 2019**).
- Phosphorus: was calculated using the **Chapman and Pratt (1961)** technique. The percentage of fruit oil (%) according to **Banat et al. (2013)**, the Soxhlet oil extraction apparatus with a boiling point of 60–80°C was used to determine the oil percentage in the fruit flesh on a dry weight basis.
Total carbohydrates (%) were calculated using the **Dubois et al. (1956)** method in dried samples of leaves and shoots.

Statistical analysis:

According to **Ott and Longnecker (2015)**, a one-way ANOVA was performed on the collected data, and CoSat CoHort Software (2005, Pacific Grove, CA, USA) was utilized to compare the means of the treatments using the least significant difference (LSD) at 0.05% (**Snedecor and Cochran, 1990**).

RESULTS AND DISCUSSIONS

A) Vegetative growth

The findings shown in **Table (2)** and **Fig. (1)** showed that *Moringa oleifera* extracts and amino acids had a substantial impact on olive vegetative growth, including plant height, leaf count, leaf area index, and total chlorophyll. However, results showed that high concentration of MLAE 12 g/l + Amino acid 100 mg/l recorded the higher values of plant height (25.01 cm), number of leaves (43.35/ plant), leaf area index (6.58 cm²) and total chlorophyll (43.62 SPAD), followed by MLAE 12 g/l which recorded plant height (21.66 cm), number of leaves (40.66/ plant), leaf area index (6.26 cm²) and total chlorophyll (39.59 SPAD) and amino acid 100 mg/l recorded plant height (16.75 cm), number of leaves (37.04/ plant), leaf area index (5.3 cm²) and total chlorophyll (38.42 SPAD), as compared to control treatment which recorded the lower values of plant height (12.31 cm), number of leaves (28.92/ plant), leaf area index (4.26 cm²) and total chlorophyll (31.54 SPAD), respectively.

Amino acids' beneficial effects on olive tree vegetative growth may be explained by their role in protein synthesis and the production of enzymes crucial to metabolism (**Mervat et al., 2015**). Furthermore, according to **Feng et al. (2020)**, amino acids' buffering properties and metabolic functions aid in controlling cellular pH. Maintaining an environment that supports enzymatic reactions and cellular metabolism depends on this regulation. Furthermore, amino acids provide plant cells with an immediate source of nitrogen and can also act as a source of carbon and energy in the event that the plant has a shortage of carbohydrates (**Zayed et al., 2023**).

The results are consistent with those reported by **El Badawy (2019)** on apricot and **Psarras *et al.* (2024)** on olive. **Nafea and Al-Janabi (2021)** demonstrated how utilizing amino acids greatly enhanced the length of olive shoots. Furthermore, **El-Bolok and Kasem (2023)** found that spraying olive trees cv. "Agazzi" with 1% chitosan and 2% amino acids improved their vegetative growth. This implies that glutamic acid is essential for enhancing plant metabolism and nutrient absorption, which results in increased growth and productivity (**Shehata *et al.*, 2021**). This season's production boost was probably caused by glutamic acid, which is known to function as a precursor for other amino acids and has been demonstrated to enhance plants' ability to absorb nitrogen efficiently (**Khalil and Hussein, 2020**).

These findings show that amino acids have a beneficial effect on fruit development most likely because they are involved in a number of physiological processes such protein synthesis, food absorption, and enzyme activation (**Shehata *et al.*, 2021**). It is well recognized that amino acids raise a plant's general vigor, which can directly lead to larger and better-quality fruits. By increasing the activity of enzymes involved in lipid production, amino acids encourage the buildup of oil in fruits. These amino acids are also known to promote photosynthetic efficiency, carbon fixation, and chlorophyll content all of which lead to higher oil content and better fruit quality (**Abdel Mawgoud *et al.*, 2020**). Glutamic acid's ability to increase fruit production is consistent with studies showing glutamic acid is important for plant growth and development. It has been demonstrated that glutamic acid enhances a number of physiological processes in plants, which can increase fruit qualities (**Eisaa *et al.*, 2023**). According to **Liu *et al.* (2020)**, the amino acid mix's better performance in the second season corroborates research showing that combining amino acids can improve plant growth and fruit yield in a synergistic way. This outcome validates glutamic's function in encouraging fruit production. By improving water retention, lowering oxidative stress, and boosting antioxidant enzyme activity, glutamic acid increases plant tolerance. Additionally, it increases proline levels for osmotic equilibrium and ABA buildup, which activates drought-resistance pathways that fortify cellular defenses and lessen damage under stress (**Xu *et al.*, 2020**). These actions may result in a higher yield of fruit. Additionally, in the second season, the amino acid mix which contains glutamic acid, proline, and gapaachieved the highest fruit output (38.75 fruits per meter). This result is in line with research indicating that amino acid combinations may have synergistic effects on fruit output and plant growth (**Liu *et al.*, 2022**).

These results also follow the same pattern as those of **Sardar *et al.* (2021)**, who reported that moringa extract has a high concentration of hormones, macro and microminerals, amino acids in the proper quantities, and many other beneficial substances. Additionally, they noted that it significantly enhanced plant development and that stevia leaves had a high concentration of chlorophyll pigment. Because moringa extracts contain growth-promoting compounds like vitamins that are essential for the digestion of carbohydrates, amino acids, and other biochemical processes, they improved growth features. Additionally, the moringa extract has significant levels of proteins and carbohydrates, which give plants the energy they need to grow and develop (**Taiz and Zeiger, 2010**). Additionally, photosynthesis, respiration, carbohydrate metabolism, enzymatic activation, and other metabolic processes that greatly contribute to cell division and elongation are all found in the aforementioned extract. These processes are

positively correlated with the traits of vegetative growth (**Marschner, 2012**). The current findings about the effects of foliar application of algae and moringa leaf extracts on vegetative growth, number of leaves per shoot, leaf area, and its mineral content are consistent with those of **Abd El Moniem and Abd Allah, (2008)**, who discovered that the increases in vegetative growth characteristics brought about by algae extracts may be due to the role of algae extracts as biostimulants for plant growth and development due to the presence of trace elements, organic substances like amino acids, auxin, and gibberellins, which enhanced vegetative growth and nutritional status.

The obtained results, however, are consistent with those of **Howladar (2014); Rady et al. (2015)**, who discovered that foliar sprays of moringa leaf extract, which is regarded as a natural plant growth stimulant, may be responsible for the improvement in vegetative growth and leaf mineral content. Zeatin, a natural derivative of cytokinin, vitamin E, proteins, ascorbates, phenols, several mineral elements, and essential amino acids. According to additional investigations, moringa extract functions as a plant hormone that promotes crop growth (**Rehman et al., 2014**).

The obtained results are consistent with those of **Thomas and Howarth (2000)**, who demonstrated that the application of MLE extract, which is rich in zeatin-like cytokinin and may induce cytokinin biosynthesis, resulted in the maximum number of photosynthetic active leaves. This is evident from the number and area of leaves per plant, which maintains higher concentrations of chlorophylls and affects plant yield. Additionally, foliar application of moringa that includes adequate amounts of stimulant compounds promotes increased cell division rate, cell enlargement, plant strength, and ultimately increases production (**Fuglie, 2000**). Additionally, **Nagar et al. (2006)** noted that zeatin hormone, which promotes plant development, is abundant in moringa leaves. Furthermore, these findings are consistent with those of **Ahmed et al. (2014); Adel (2014)** found that spraying mango trees with algae extracts either by themselves or in combination with natural plant extracts was highly successful in increasing fruit set, fruit retention, yield, and improved fruit quality. Additionally, the same findings are consistent with those of **Hafez et al. (2013)** on olives and **Hegab et al. (2005)** on citrus.

These results were consistent with those of **Al-Sabbagh et al. (2020)**, who found that foliar spraying Washington navel oranges with moringa extracts significantly increased the number of shoots, leaves, and leaf area. **Al-Abed-Allah (2022)** also noted that Mexican lime transplants treated with moringa extracts showed a significant rise in the quantity and area of shoots and leaves. In addition to being utilized to boost fruit quality and productivity, moringa leaf extract is not a substitute for inorganic fertilizers (**Phiri, 2010**). Moringa leaf extract has received a lot of interest lately due to its high cytokinin concentration (**Abdalla, 2013**). Because moringa leaf extract is high in nutrients like N, P, K, and micronutrients and aids in their absorption, it may promote both root length and vegetative development. Additionally, the same scientists noted that moringa leaf extract can boost the tree's strength due to its high antioxidant and phytochemical content (**Rani et al., 2018; and Meireles et al., 2020**). According to **Semida et al. (2014)**, foliar spraying MLE can help plants develop and produce by acting as a biostimulant. Leaf moringa has a significant impact on growth and production since it contains amino acids, minerals including calcium, magnesium, phosphorus, sulfur, zinc, iron, and copper, as well as vitamin E (**Rady et al., 2015; Nisar et al., 2021**). The nutritional components that moringa extracts contain, such as magnesium, which is

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involved in the structure of the chlorophyll molecule, iron, which aids in its formation, and copper, which stabilizes and prevents its degradation, may have an impact on how much chlorophyll is present (Marschner, 2012).

Table (2): Impact of *Moringa oleifera* extracted and amino acid on shoot length, number of leaves/ tree, leaf area and total chlorophyll of olive tree cv. "Koroneiki" during 2025 season.

Treatments	Shoot length (cm)	No. of leaves/ tree	LA (cm ²)	Total chlorophyll (SPAD)
Control	12.31	28.92	4.26	31.54
MLAE 4 g/l	18.53	38.71	5.27	35.69
MLAE 8 g/l	20.75	39.24	6.01	37.39
MLAE 12 g/l	21.66	40.66	6.26	39.59
Amino acid 50 mg/l	13.93	31.68	4.63	34.33
Amino acid 75 mg/l	15.06	34.01	5.05	36.05
Amino acid 100 mg/l	16.75	37.04	5.35	38.42
MLAE 4 g/l + Amino acid 50 mg/l	22.47	40.85	6.43	41.29
MLAE 8 g/l + Amino acid 75 mg/l	23.54	42.28	6.51	41.88
MLAE 12 g/l + Amino acid 100 mg/l	25.01	43.35	6.58	43.62
LSD(0.05)	1.47	0.19	0.05	1.43

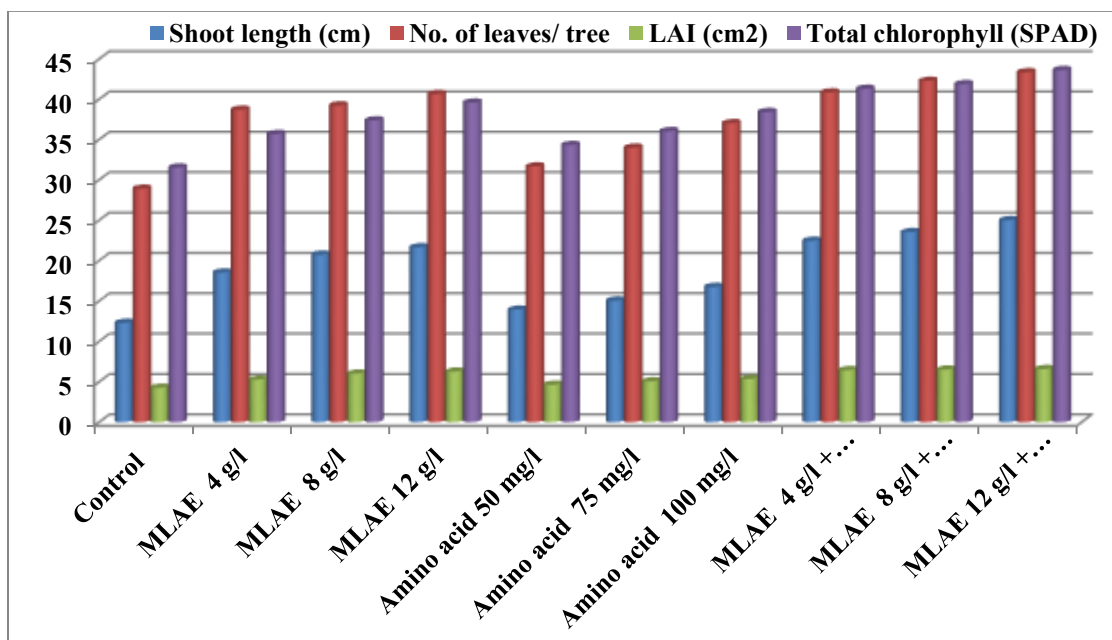


Fig. (1): Impact of *Moringa oleifera* extracted and amino acid on shoot length, number of leaves/ tree, leaf area and total chlorophyll of olive tree cv. "Koroneiki" during 2025 season.

B) Yield and yield components

The results shown in **Table (3)** and **Fig. (2)** showed that the fruit length, fruit diameter, fruit weight, flesh weight, and yield yield/tree of olive cv. "Koroneiki" were all significantly impacted by *Moringa oleifera* extracts and amino acids. However, results showed that, mixture of *Moringa oleifera* extracts and amino acid at concentration MLAE 12 g/l + Amino acid 100 mg/l recorded the higher values of fruit length (2.84 cm), fruit diameter (2.05 cm), fruit weight (5.29 cm), flesh weight (4.55 g) and fruit yield (32.87 kg/ tree), followed by MLAE 12 g/l which recorded fruit length (2.15 cm), fruit diameter (1.34 cm), fruit weight (3.51 cm), flesh weight (3.03 g) and fruit yield (28.03 kg/ tree) and amino acid 100 mg/l recorded fruit length (2.40 cm), fruit diameter (1.72 cm), fruit weight (4.39 cm), flesh weight (2.86 g) and fruit yield (29.90 kg/ tree), as compared to control treatment which recorded the lower values of fruit length (1.60 cm), fruit diameter (1.16 cm), fruit weight (2.99 cm), flesh weight (1.99 g) and fruit yield (23.32 kg/ tree), respectively.

This implies that glutamic acid is essential for enhancing plant metabolism and nutrient absorption, which results in increased growth and productivity (**Shehata et al., 2021**). This season's production boost was probably caused by glutamic acid, which is known to function as a precursor for other amino acids and has been demonstrated to enhance plants' ability to absorb nitrogen efficiently (**Khalil and Hussein, 2020**). These findings show that amino acids have a beneficial effect on fruit development, most likely because they are involved in a number of physiological processes such protein synthesis, food absorption, and enzyme activation (**Shehata et al., 2021**). It is well recognized that amino acids raise a plant's general vigor, which can directly lead to larger and better-quality fruits. These results are consistent with other studies showing how important amino acids are for fruit development, oil generation, and plant growth. As the building blocks of proteins, amino acids are vital to plant metabolism. They take part in the synthesis of hormones and enzymes that control a number of physiological functions, such as the creation of secondary metabolites, stress tolerance, and nutrition uptake (**Shehata et al., 2021**). In particular, amino acids like proline, glutamic acid, and others can improve the plant's ability to absorb nutrients and generate metabolites, which are essential for olive tree oil synthesis (**Khalil and Hussein, 2020**). By increasing the activity of enzymes involved in lipid production, amino acids encourage the buildup of oil in fruits. These amino acids are also known to promote photosynthetic efficiency, carbon fixation, and chlorophyll content all of which lead to higher oil content and better fruit quality (**Abdel Mawgoud et al., 2020**).

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Table (3): Impact of *Moringa oleifera* extracted and amino acid on fruit length, diameter, weight, flesh weight and yield/tree of olive tree cv. "Koroneiki" during 2025 season.

Treatments	Fruit length (cm)	Fruit diameter (cm)	Fruit weight (g)	Flesh weight (g)	Yield (kg/tree)
Control	1.60	1.16	2.99	1.99	23.32
MLAE 4 g/l	1.84	1.23	3.23	2.40	25.27
MLAE 8 g/l	1.98	1.27	3.35	2.66	26.70
MLAE 12 g/l	2.15	1.34	3.51	3.03	28.03
Amino acid 50 mg/l	2.28	1.45	3.85	3.41	28.38
Amino acid 75 mg/l	2.32	1.54	4.18	3.70	29.39
Amino acid 100 mg/l	2.40	1.72	4.39	2.86	29.90
MLAE 4 g/l + Amino acid 50 mg/l	2.56	1.84	4.66	4.19	30.59
MLAE 8 g/l + Amino acid 75 mg/l	2.74	2.00	5.00	4.36	31.64
MLAE 12 g/l + Amino acid 100 mg/l	2.84	2.05	5.29	4.55	32.87
LSD_(0.05)	1.83	1.77	5.22	3.90	10.44

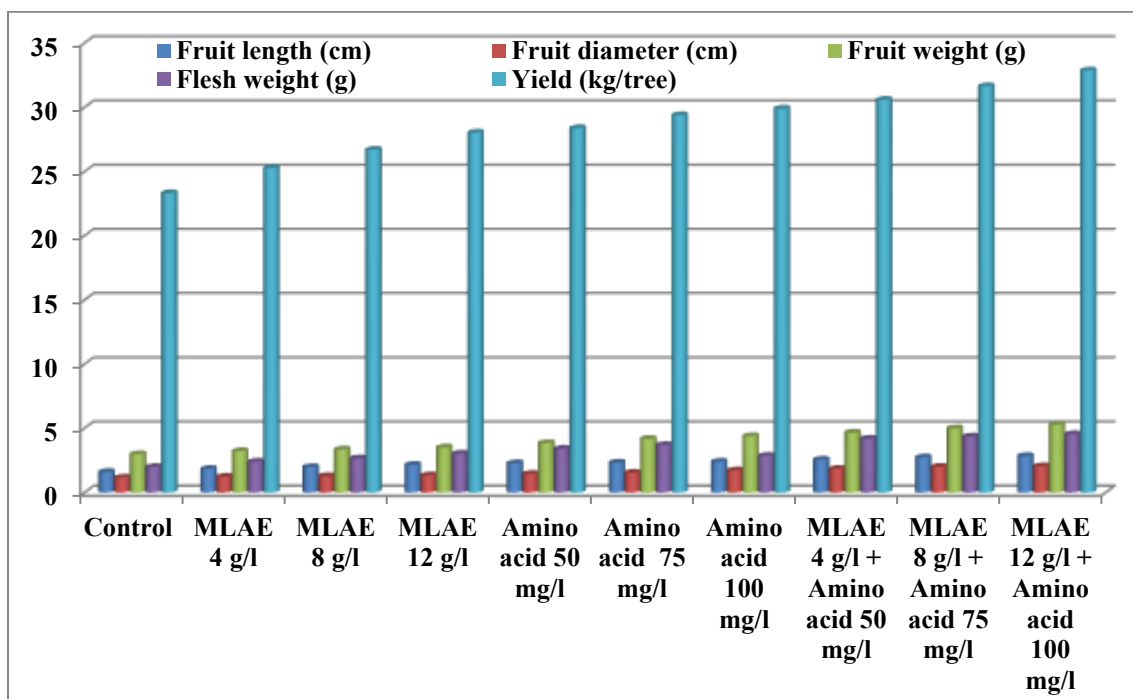


Fig. (2): Impact of *Moringa oleifera* extracted and amino acid on fruit length, diameter, weight, flesh weight and yield/tree of olive tree cv. "Koroneiki" during 2025 season.

C) Chemical composition

The results shown in **Table (4)** and **Fig. (3)** showed that the nitrogen, phosphorus, potassium, protein, total carbohydrate, and oil content of olive cv. "Koroneiki" were considerably impacted by *Moringa oleifera* extracts and amino acids. However, results showed that, mixture of *Moringa oleifera* extracts and amino acid at concentration MLAE 12 g/l + Amino acid 100 mg/l recorded the higher values of nitrogen (2.40%), phosphorus (0.89%), potassium (2.55 %), protein (15.00 %), total carbohydrate (35.19%) and oil content (36.02 %), followed by MLAE 12 g/l which recorded nitrogen (1.72 %), phosphorus (0.80 %), potassium (2.33 %), protein (10.75 %), total carbohydrate (26.44%) and oil content (33.40%) and amino acid 100 mg/l recorded nitrogen (1.66 %), phosphorus (0.67 %), potassium (1.94 %), protein (10.36 %), total carbohydrate (20.62%) and oil content (29.75 %), as compared to control treatment which recorded the lower values of nitrogen (1.18 %), phosphorus (0.42 %), potassium (1.26 %), protein (7.40 %), total carbohydrate (16.02%) and oil content (22.58 %), respectively.

According to **Abd El-Hamied and El-Amary, (2015)**, spraying moringa leaf extract at 2 and 4% on pear trees of the Le-Conte variety increased the leaf mineral content of potassium, phosphorus, and nitrogen in comparison to the control. Furthermore, compared to the control, **Abd Al Rhman et al. (2018)** found that spraying moringa leaf extract at 0.50 ml/L on orange cultivars "Washington" and "Murcott Tangor" greatly enhanced the leaf mineral composition from nitrogen, phosphorous, potassium, zinc, iron, and manganese. The impact of amino acids on raising protein content is interpreted differently by different researchers. According to some writers, plants prefer to absorb amino acids in reduced nitrogen forms since they need less energy for protein synthesis (**Günes et al., 1996**). According to **Liu et al. (2014)**; **Mobini et al. (2014)**, the impact of amino acids on nitrate metabolism enzymes, such as nitrate reductase and glutamine synthetase, is the primary cause of the effect of amino acids on nitrogen assimilation and protein synthesis.

Because the extract from moringa leaves contains macronutrients, particularly N, P, and K, as well as micronutrients that are directly absorbed when sprayed on the leaves, increasing their percentage in the leaves, it has the effect of increasing the content of macro elements and total phenols in the leaves. According to the extract's function in enhancing the features of vegetative development, which initiates the process of nutrient absorption and raises the concentration of nutrients in the leaves. These findings support the findings of (**Abo El-Enien et al., 2015**) on navel orange trees. Additionally, **Mohamed et al. (2017)** found that foliar application of moringa leaf extracts significantly increased the leaf content of phosphorus, potassium, and total chlorophyll in four kinds of citrus rootstocks.

In particular, amino acids like proline, glutamic acid, and others can improve the plant's ability to absorb nutrients and generate metabolites, which are essential for olive tree oil synthesis (**Khalil and Hussein 2020**). By increasing the activity of enzymes

involved in lipid production, amino acids encourage the buildup of oil in fruits. These amino acids are also known to promote photosynthetic efficiency, carbon fixation, and chlorophyll content all of which lead to higher oil content and better fruit quality (**AbdelMawgoud *et al.*, 2020**).

When plants are subjected to biotic and abiotic stressors, amino acids have been shown to have a biostimulatory effect. These acids contribute to the plants' increased overall protein composition. The higher concentration of ribosomal proteins and their function as the structural component of proteins may be connected to the higher capacity of total proteins. Additionally, as organic protein molecules and the building blocks of proteins, amino acids can be directly absorbed by plant cells (**Abd El-Aziz *et al.*, 2009**). By enhancing their ability to withstand environmental challenges and boosting their protein and chlorophyll levels, amino acids have an impact on plant growth and yields. Glutamic acid influences stomatal opening and closure as an osmotic agent in the cytoplasm of stomatal cells. Additionally, arginine promotes the production of plant hormones that affect fruit production and flowering (**Pouryousef and Shahravan, 2014**). Amino acids have both direct and indirect effects on plant growth and development as well as physiological processes (**Faten *et al.*, 2010**).

By raising their nitrogen content, the use of substances containing free amino acids proved successful in raising the protein content of peas. This study shown that applying amino acids topically greatly raised the N, P, and K contents of peas (**Mahmoudi, 2012**). The oil content of "Arbequina" can be significantly increased by applying the amino acid glutamine. The cultivar has the greatest influence on the final quality of olive oil and fruits. The cultivar affects the properties and content of olive oil as well as the effectiveness of oil extraction (**Fedeli, 1977**). According to **Hamidoghli *et al.* (2008)**, oil content is cultivar-dependent, meaning that the three olive cultivars of "Roghani," "Zard," and "Lechino" had considerably different oil percentages. According to **Najafian *et al.* (2008)**, cultivar treatment significantly affected oil content, with "Koroneiki" and "Mission" having the highest and lowest oil concentrations, respectively, at 71.3% and 63.8%. According to **Jami *et al.* (2016)**, "Koroneiki" had the lowest oil content while "Amygdalolia" had the greatest. The protein composition of olive cultivars and plant sections varies significantly, according to **Lazovic *et al.* (1999)**. They found that the fruit had a protein level of 1.5–2.61% and the protein content of olive buds ranged from 4.72 to 6.74%. Olive buds had the highest protein concentration, followed by leaves, while fruits had the lowest protein amount. This report is supported by our findings. According to a field study, using substances that contain free amino acids can boost the protein content of pea grains by raising their nitrogen percentage (**Mahmoudi, 2012**). According to **Liu *et al.* (2014)**; **Mobini *et al.* (2014)**, the impact of amino acids on nitrate metabolism enzymes, such as nitrate reductase and glutamine synthetase, is the primary mechanism by which amino acids influence nitrogen assimilation and protein synthesis.

Impact of *Moringa oleifera* extracted and amino acid on growth and quality of olive tree (*Olea europaea*)

Table (4): Impact of *Moringa oleifera* extracted and amino acid on N, P, K, protein, total carbohydrate and oil content of olive tree cv. "Koroneiki" during 2025 season.

Treatments	N (%)	P (%)	K (%)	Protein (%)	Total carbohydrate (%)	Oil content (%)
Control	1.18	0.42	1.26	7.40	16.02	22.58
MLAE 4 g/l	1.41	0.61	1.93	8.82	20.32	29.22
MLAE 8 g/l	1.57	0.67	2.17	9.79	22.46	32.69
MLAE 12 g/l	1.72	0.80	2.33	10.75	26.44	33.40
Amino acid 50 mg/l	1.36	0.50	1.61	8.50	17.56	24.78
Amino acid 75 mg/l	1.52	0.56	1.81	9.53	20.22	25.96
Amino acid 100 mg/l	1.66	0.67	1.94	10.36	20.62	29.75
MLAE 4 g/l + Amino acid 50 mg/l	1.97	0.67	2.12	12.30	29.11	30.48
MLAE 8 g/l + Amino acid 75 mg/l	2.21	0.73	2.39	13.84	33.66	33.86
MLAE 12 g/l + Amino acid 100 mg/l	2.40	0.89	2.55	15.00	35.19	36.02
LSD_(0.05)	0.04	0.03	0.03	0.26	10.21	11.52

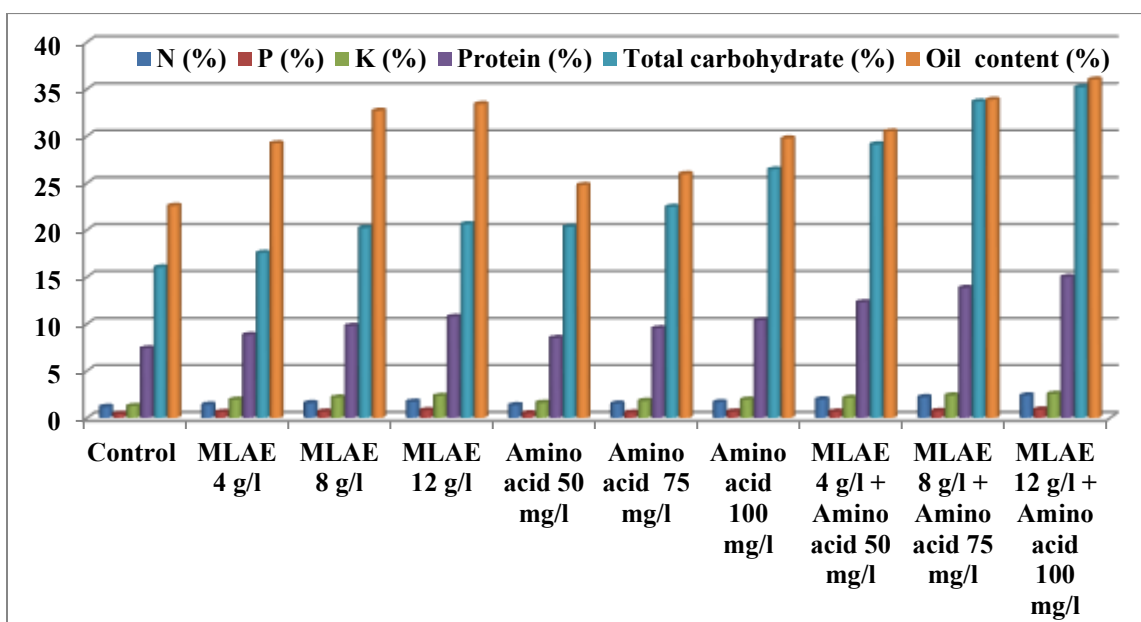


Fig. (3): Impact of *Moringa oleifera* extracted and amino acid on N, P, K, protein, total carbohydrate and oil content of olive tree cv. "Koroneiki" during 2025 season.

Conclusion:

According to the aforementioned findings, biostimulants have received a lot of interest lately as growth and yield boosters. Applying moringa extract to plants is thought to be an environmentally friendly method to improve plant growth and increase yield and fruit chemical properties. This is because such biostimulants can increase plant vigor and

yield parameters by altering plant physiological reactions and increasing plant nutrient acquisition. In contrast to the control of olive trees cv. "Koroneiki," it could be determined that all spraying treatments at various concentrations and dates had a favorable influence on enhancing vegetative growth, leaf mineral contents, yield, and improving fruit chemical qualities.

It is possible to draw the conclusion that foliar spraying of amino acids improves the nutritional status of olive trees and produces a high yield with good fruit quality under similar conditions based on the current findings. Applying amino acids has a direct or indirect impact on the physiological processes involved in plant growth and development. They are crucial for promoting cell growth and are regarded as precursor components of proteins. Additionally, amino acids have been applied topically to control fruit quality, yield, and vegetative growth. Amino acids play a significant role in the growth and nutritional health of olive trees, and their beneficial effects on physiological activity stimulation are undoubtedly reflected in the promotion of growth, flowering, and fruit quality.

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